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School of Science Engineering and Technology Skills

Department of Engineering

Subject – AWS Lab

Experiment - 2

Aim: Creation of IAM Users, Groups, Roles, and Policies

Lab Objective:

Create a secure AWS IAM setup following the principle of least privilege, where:

- A Junior Developer has read-only access to Amazon S3 buckets (no upload or delete permissions).
- An EC2 Admin role is available for automated services that require full S3 access.

Step 1: Create a Custom IAM Policy

Instead of granting full administrative access, we create a custom policy with limited permissions.

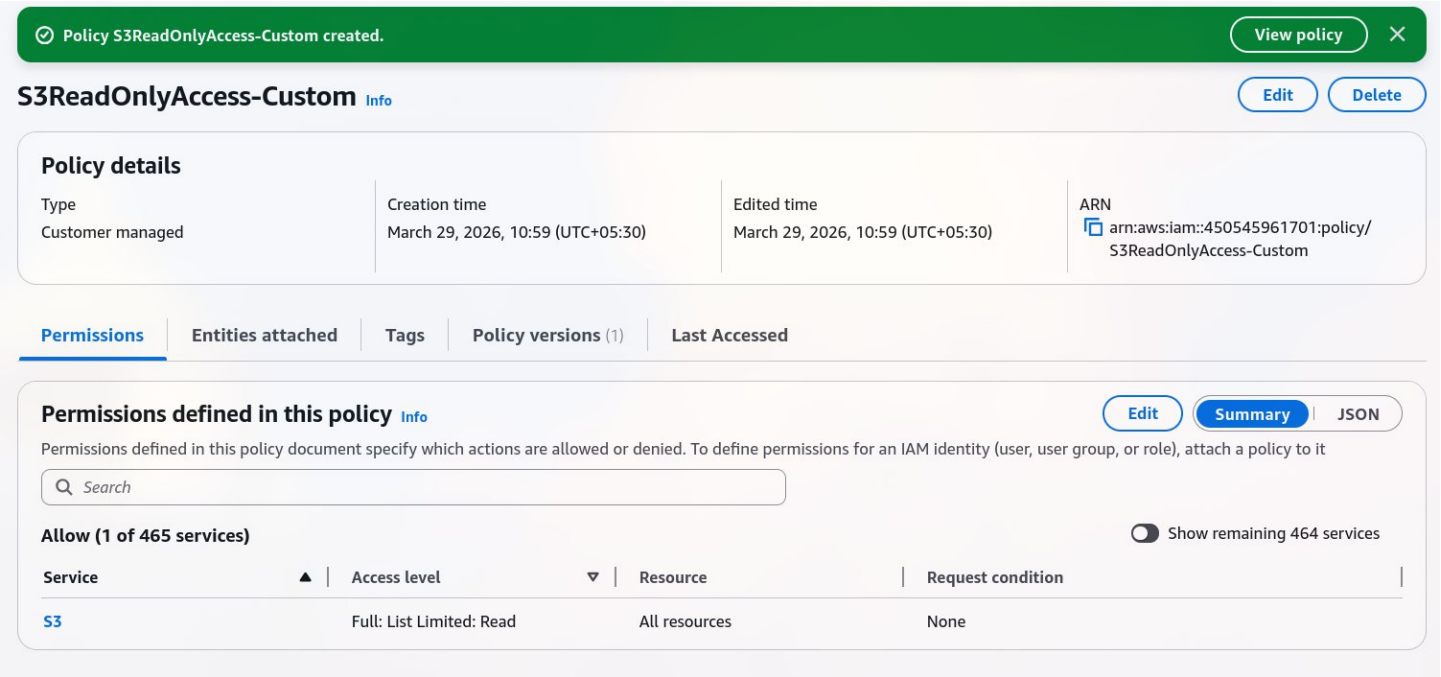
Steps:

1. Sign in to the AWS Management Console.
2. Search for and open IAM.
3. In the left navigation pane, click Policies → Create policy.
4. Select the JSON tab.
5. Paste the following policy:

```
{
  "Version": "2012-10-17",
```

```
        "Statement": [
            {
                "Effect": "Allow",
                "Action": [
                    "s3:Get*",
                    "s3:List*"
                ],
                "Resource": "*"
            }
        ]
    }
```

6. Click Next.
7. Policy name: S3ReadOnlyAccess-Custom.
8. Click Create policy.

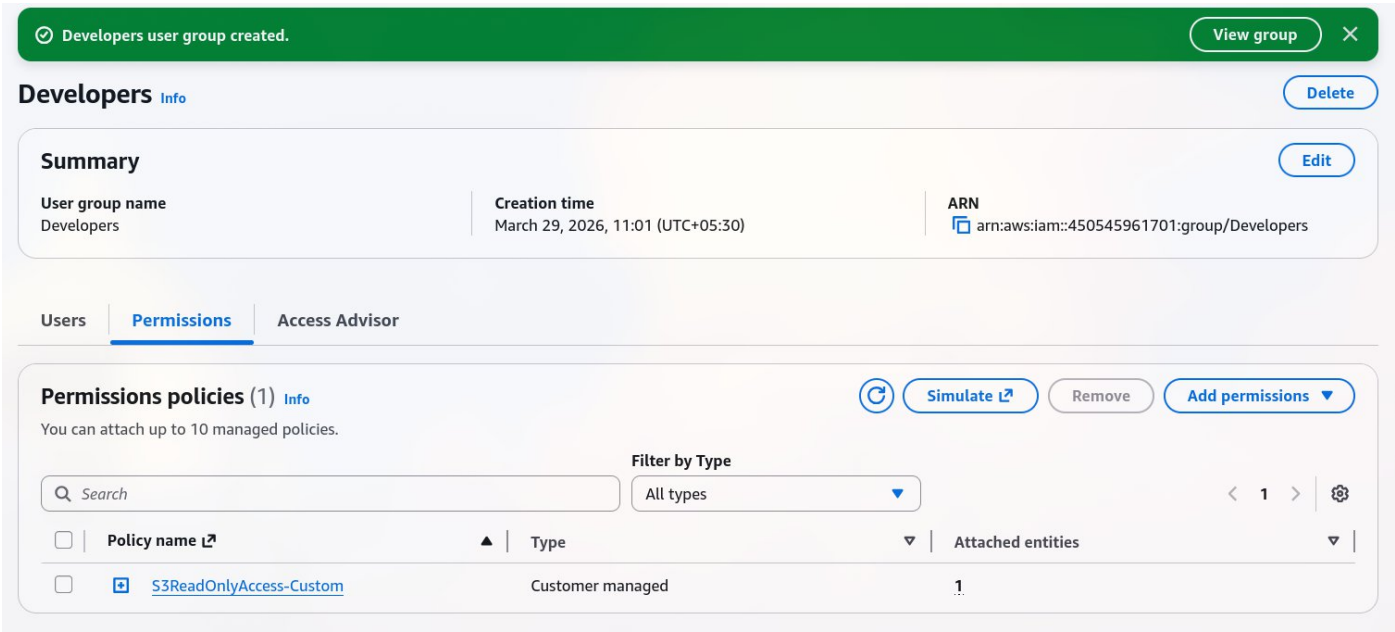


Step 2: Create an IAM Group

Groups simplify permission management for multiple users.

Steps:

1. In the IAM dashboard, click User groups → Create group.
2. Group name: Developers.
3. Under Attach permissions policies:
    - Search for S3ReadOnlyAccess-Custom.
    - Select the policy.
4. Click Create group.



Step 3: Create an IAM User

This user represents a Junior Developer.

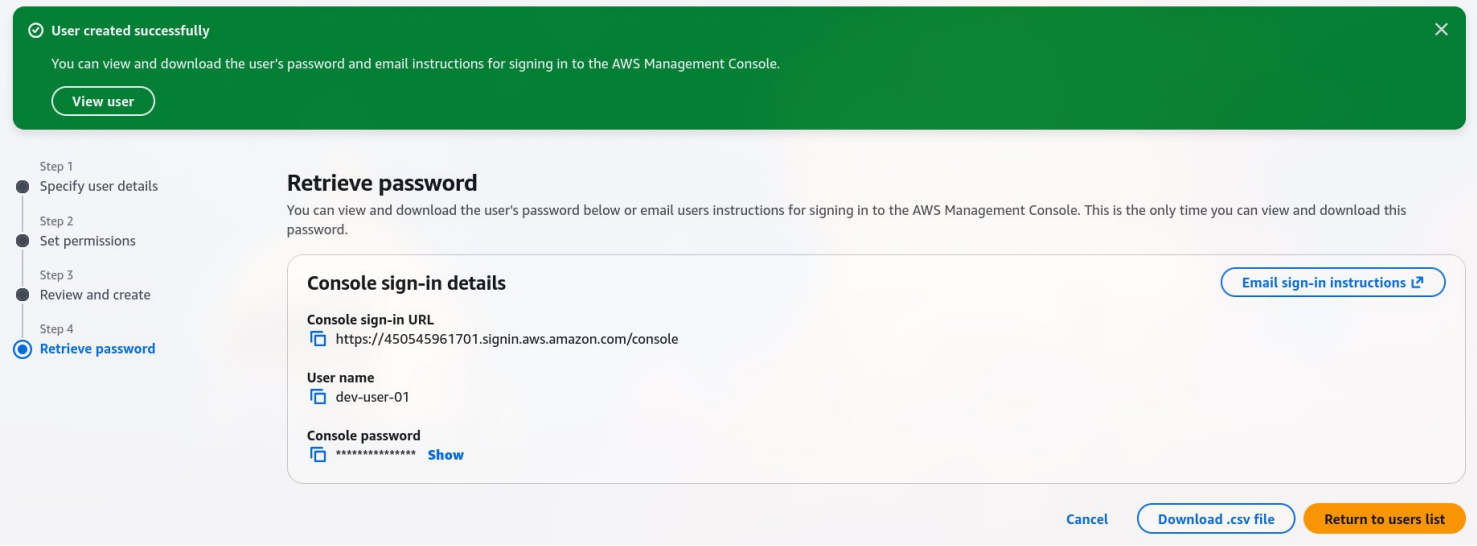
Steps

1. Click Users → Create user.
2. User name: dev-user-01.
3. Enable Provide user access to the AWS Management Console.
4. Select I want to create an IAM user.
5. Set a Custom password.
6. Click Next.

7. On the Set permissions page:
    - Choose Add user to group.
    - Select the Developers group.
8. Click Next → Create user.

Important: Download the .csv file after creation. It contains:

- Console sign-in URL
- Username
- Password



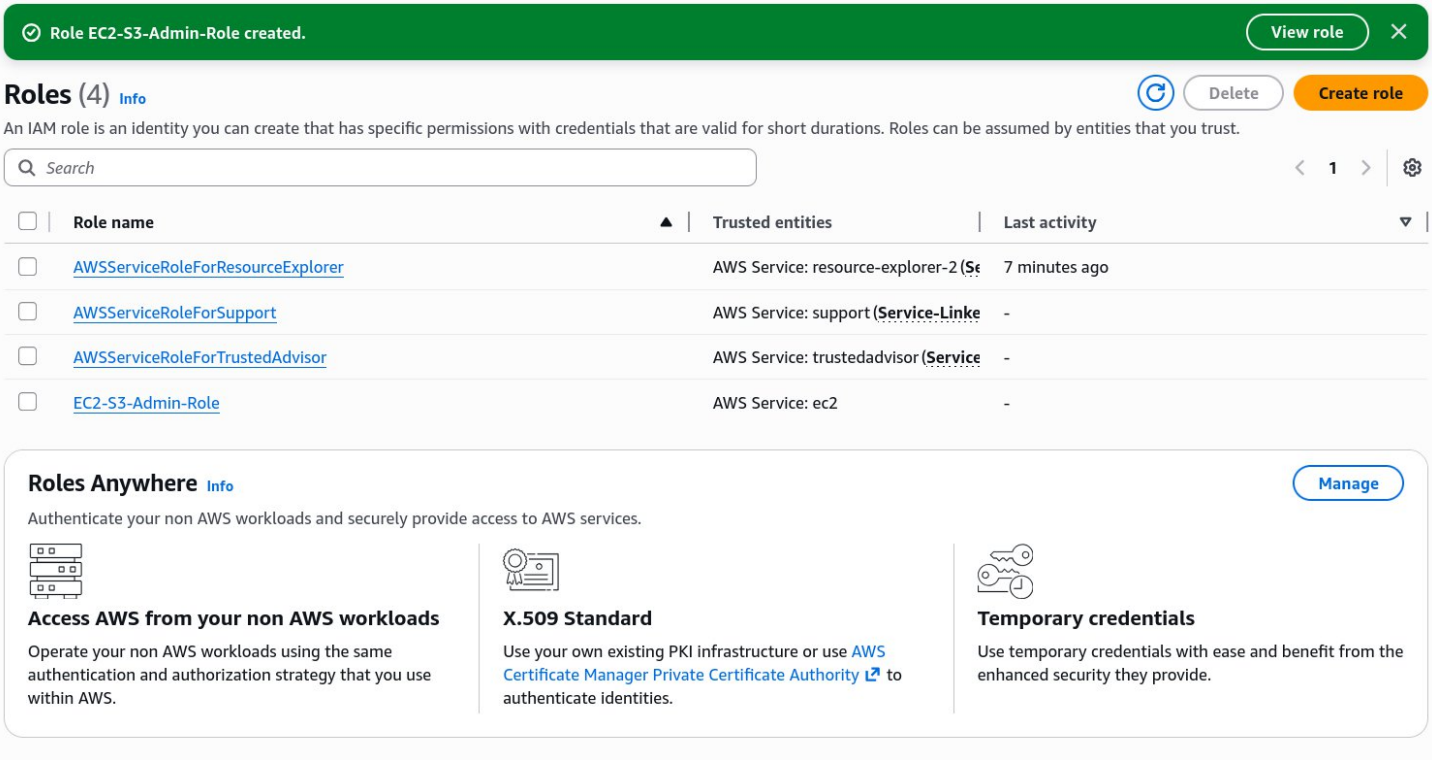
Step 4: Create an IAM Role (For AWS Services)

IAM roles allow AWS services (such as EC2) to access other AWS resources securely without passwords.

Steps:

1. Click Roles → Create role.
2. Trusted entity type: AWS service.
3. Use case: EC2.
4. Click Next.
5. Under Add permissions:
    - Search for AmazonS3FullAccess (used for this lab).
    - Select the policy.
6. Click Next.

7. Role name: EC2-S3-Admin-Role.
8. Click Create role.



Verification Task:

Confirm that permissions are working as expected.

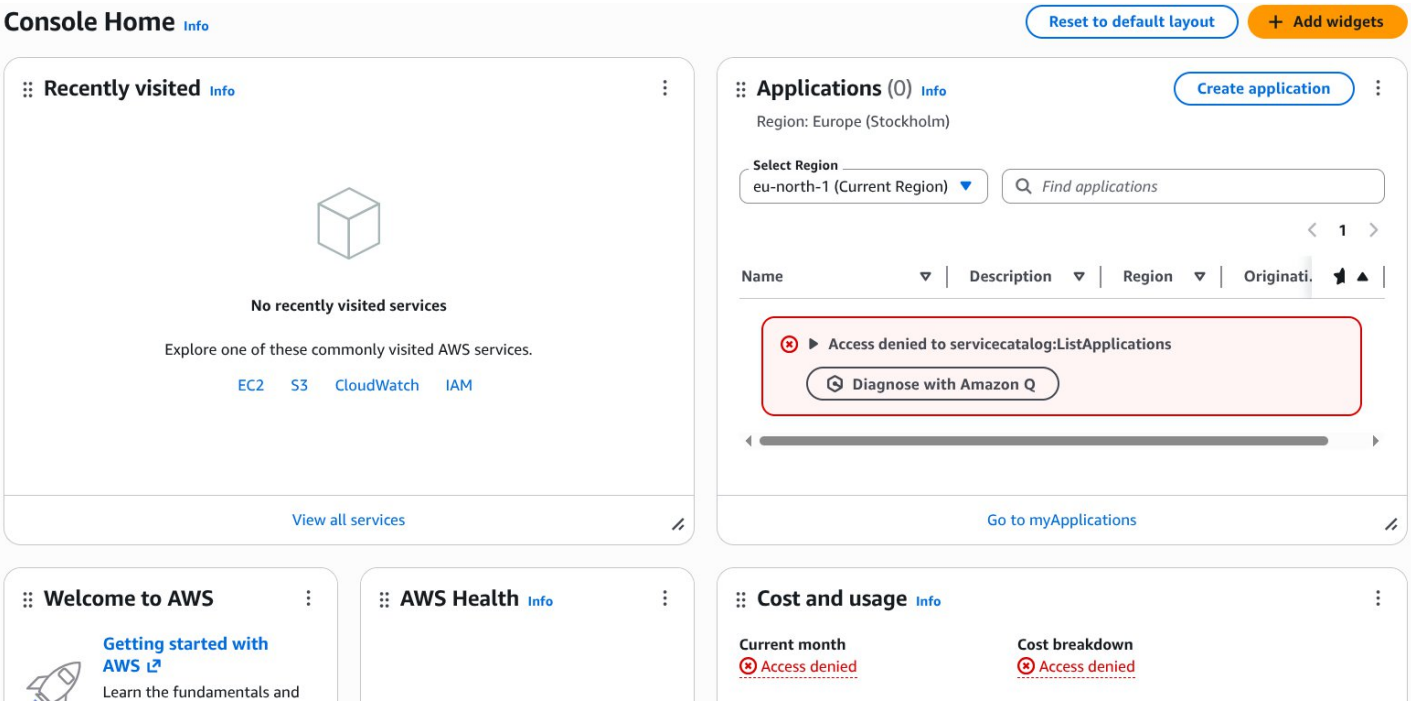
Steps:

1. Open a Private / Incognito browser window.
2. Use the Console sign-in URL from the downloaded CSV file.
3. Log in as dev-user-01.
4. Navigate to Amazon S3.

Expected Results:

- You can view and list S3 buckets.
- You cannot upload files.
- You cannot delete buckets or objects.

An “Access Denied” error confirms that the least-privilege policy is working correctly.



Outcome:

You have successfully:

- Implemented a custom IAM policy
- Used IAM groups for permission management
- Created a restricted IAM user
- Configured an EC2 service role for automated access

This setup improves security while maintaining operational flexibility.